

5(a)(i)	<i>'if qualified' = ref. to reduced, gas exchange/diffusion</i> <i>idea that</i> (ciliated epithelial cells), not thin enough / too thick / not flat / AW ; A thicker / thick, if qualified <i>plus one from:</i> <i>context is movement of, O₂/ CO₂/ respiratory gases, across cells</i> <u>diffusion</u> too slow ; AW A slow / slower, if qualified <u>diffusion</u> distance too great ; A increases / increased, if qualified cilia, action / AW, may affect efficiency of gas exchange ; A examples e.g. may use up oxygen (so less available for gas exchange) may affect role of surfactant	2
5(a)(ii)	<i>any two from:</i> mucus, wafted / moved, qualified ; e.g. to / towards, mouth / (back of) throat away from, alveoli / gas exchange surface / lungs out of the air ways / upwards AW for <i>idea of</i> moving to back of throat I 'out of the body' / 'to the stomach' R if implied cilia are in the gut (for movement of mucus, cilia show) synchronous rhythm / metachronal rhythm ; A described e.g. all move at once / wavelike action otherwise mucus (with pathogens) accumulates / AW, with consequence ; e.g. increased risk of infection AW reduced flow of air AW no ora for this mp	2

Question	Answer	Marks
5(b)	any four from: allow O₂ for, oxygen / oxygen molecules I ref. to damage to gas exchange system as the person is stated as healthy suggestion that 1 haemoglobin has higher affinity for, pollutants / named, than oxygen; A haemoglobin has decreased affinity for oxygen 2 pollutants (may) <u>bind</u> to haemoglobin, with consequence ; e.g. so, fewer oxygen molecules bind / less oxygen binds fewer Hb binding sites available, for oxygen molecules / oxygen changes Hb shape / affects (R-group) bonds, so oxygen binding hindered I ref. to carbon dioxide 3 (pollutants) lower, proportion / partial pressure / concentration, of oxygen in (atmospheric / alveolar) air, so less, diffuses to blood / diffuses into capillary / enters red blood cells / binds to haemoglobin ; 4 polluted air leads to <u>less steep, diffusion / concentration, gradient</u> between, alveolar air / alveolus, and, blood / capillary); A smaller difference in concentration 5 pollutant, presence / binding, affects, haemoglobin allostery / cooperative binding / described ; ora e.g. because without pollutant the first O₂ binding to Hb makes it easier to bind the next O₂	4